

ABOM and OINL for Anonymity Claims: Proof-Carrying Odds–Inflation Budgets Without Publishing Full Artifacts

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Abstract

This series publishes brief, referee-grade papers, but a credible anonymity claim typically depends on a large artifact surface: measurement scripts, traces, compiler knobs, certificates, and verifier outputs. Publicly shipping *all* artifacts is often infeasible (space, privacy, operational constraints).

We propose two lightweight documentation objects that make claims checkable even in a source-only release. An *Anonymity Bill of Materials* (ABOM) is a signed, machine-checkable manifest that commits (by digest) to the artifacts and threat-model assumptions behind a claim. An *Odds-Inflation Nutrition Label* (OINL) is a human-readable summary extracted from the ABOM: it states the protected interface (ENF-ID), the per-epoch budget representation, and the intended window/threat declaration (TW-ID). Together they support reproducible *interfaces*: others can diff labels across releases, verify signatures and digests, and request the full artifact archive only when needed.

Scope. We isolate and formalize *ABOM and OINL for Anonymity Claims* as a reusable contract surface in the anonymity / Anonymous-DHT series.

Imports. We treat odds inflation and endpoint sizing as imported (Anonymity A; Synthesis 5).[2, 3] We adopt the contract-object vocabulary (Synthesis 0) as the schema surface to which ABOM commits; CPPC manifests are a control-plane sibling of ABOM (AnonDHT State 3).[9, 13] Interface declarations (exposure normal forms / ENF-ID) and threat/window declarations (TW-ID) are imported from the trace-interface and threat-window synthesis notes.[6, 7] Replay-plan and state-declaration equivalence/drift rules are imported from Synthesis 18.[11] For supply-chain style release receipts and transparency anchoring, see Release A and W-Congestion-EQ.[4, 5]

Artifact policy. This archive is source-only; supporting artifacts (logs, traces, scripts, certificates, and verifier outputs) are available on request as a single bundle, committed by ABOM/OINL manifests (this paper) and governed by the series artifact policy (Synthesis 6).[1]

Exports. Exports two schemas: (i) *ABOM* (signed manifest of claim semantics and artifact digests) and (ii) *OINL* (a derived, human-readable label with claim id, projection, adjacency, budget representation, and horizon).

1 Design goals

ABOM borrows its “bill of materials” stance from software supply-chain practice: rather than publishing every artifact, we publish a signed manifest that commits to them and to their semantics. This parallels SBOM standards (e.g., SPDX, CycloneDX) and attestation frameworks such as in-toto, but with anonymity-specific fields (threat windows, transcript projections, and budget accountants).[14, 16, 15, 17]

ABOM/OINL are *interfaces*, not proofs. They aim to make three failure modes harder:

1. **Silent drift:** knobs or evidence change without an auditable trail.
2. **Cherry-picked evidence:** a claim is supported by a subset of runs while other runs contradict it.
3. **Ambiguous semantics:** a budget is stated in a representation that downstream readers cannot interpret.

2 ABOM: the source of truth

Definition 1 (ABOM). *ABOM is a contract object in the shared vocabulary of Synthesis 0; this note records the anonymity-specific minimum surface.[9] An ABOM is a signed manifest that commits (by digest) to: a claim identifier and version; an ENF-ID naming the protected transcript projection; a TW-ID naming the threat/window declaration; an adjacency declaration; a budget representation; replay hooks (plan identifiers/spec digests); optional state-surface declarations when guarantees depend on long-lived control-plane state; artifact digests/availability flags; and a signature.*

Minimal schema sketch. The schema is intentionally permissive; it is a contract object for auditors. A minimal ABOM (YAML/JSON) might contain:

```
{
  "abom_version": "0.3",
  "claim_id": "anondht.destination_privacy.v1",
  "paper_ref": "anonymity_series/paperB_anonymous_dht_profiling",
  "exposure_nf_id": "sha256:<ENF-ID-hex>",
  "tw_id": "sha256:<TW-ID-hex>",
  "replay_hook": {"plan_id": "plan.replay.v1", "plan_spec_id": "sha256:<PLAN-SPEC-hex>"},
  "state_decl_id": "sha256:<STATE-DECL-hex>",
  "adjacency": "change-one-destination-class",
  "budget": {"type": "odds_inflation_bits", "epsilon_bits": 0.35, "delta": 1e-6, "unit": "lookup"},
  "artifacts": [
    {"name": "verifier_report.json", "sha256": "..."},
    {"name": "trace_bundle.tar.zst", "sha256": "...", "availability": "on_request"}
  ],
  "signing": {"scheme": "ed25519", "pubkey": "...", "signature": "..."}
}
```

Remark 1 (Digests are the point). *ABOM allows a publication to be “source-only” while still being checkable: if an author later provides the private artifact archive, third parties can verify that the delivered artifacts match the published digests.*

OINL field	Meaning
Claim id	Stable string used for diffing and change control
Exposure (ENF-ID)	<code>exposure_nf_id</code> naming what part of the transcript is budgeted
Adjacency	The unit of privacy / what changes between worlds
Budget type	Odds inflation / max-divergence / TV / tradeoff curve
Per-epoch budget	(η, δ) or equivalent
Threat/window + usage	<code>tw_id</code> plus the usage/horizon assumption and accounting regime (rulebook: Synthesis)
Owned proofs	Which papers/certificates provide the proof objects
Artifact digests	Hash pointers to evidence (on request)

Table 1: OINL is a derived label: it summarizes (and links to) ABOM fields.

3 OINL: a label derived from ABOM

OINL is a compact table extracted from the ABOM. It is the object we expect most readers to consume and compare.

OINL and “referee-grade brevity.” The papers in this archive can stay short because OINL externalizes the long tail of artifacts. A referee can demand the archive; a casual reader can still interpret the claim.

4 Interfacing with receipts, certificates, and transparency

Receipt line items (Synthesis 12) are the publication-level companion of ABOM/OINL: each receipt line item repeats the same claim fields (`exposure_nf_id`, `tw_id`, and when applicable `plan_spec_id` and `state_decl_id`) so that a reader can compare the OINL against the receipt without opening private artifacts.[10]

Horizon claims are structured. If an ABOM/OINL entry quotes a horizon- n total ("per day", "for Q lookups"), it should also name the usage assumption and bind the accountant via `plan_spec_id` (and `state_decl_id` when state enters), following the receipt-accounting rulebook.[8]

A drift rule worth making explicit. If `plan_spec_id` or `state_decl_id` changes, the publication must mint a new receipt/ABOM entry that pins the new digest-bound semantics. Whether the underlying privacy claim needs recertification depends on whether the change is merely “publication-only” (re-run the verifier) or claim-affecting; we centralize the taxonomy in Synthesis 18 so other papers do not restate it.[11]

ABOM is compatible with (and intentionally defers details to) the series’ receipt/certificate pipeline:

- **Release receipts:** Release A defines a ledger bundle and a compact PVSA summary attestation.[4] ABOM can point to the bundle and include the PVSA digest.
- **Congestion certificates:** W-Congestion-EQ defines congestion-specific certificate fields and log anchoring patterns.[5] ABOM can embed the certificate digest and an inclusion-proof digest.
- **Optimization certificates:** BossFight C and the 2026-01-29 wiki post treat dual certificates as proof objects for TV privacy designs. ABOM can include the primal/dual transcript digests.

5 Takeaways

- **ABOM is the commitment:** a signed manifest that pins (by digest) the artifacts and the *semantics* of the claim (projection, adjacency, budget representation, horizon), plus any replay and state surfaces the guarantee depends on.
- **OINL is the diffable summary:** a compact label extracted from ABOM so reviewers and downstream users can compare claims across releases without unpacking the full artifact archive.
- **Source-only can still be referee-grade:** publish ABOM/OINL in the paper; ship the heavy archive on request; let third parties verify the delivered bundle matches the published digests.

6 Limitations and scope

ABOM/OINL do not prevent lying: a publisher could sign a manifest for a claim that is false. Their purpose is narrower: make claims *auditable* and *comparable* across releases, and make it straightforward for others to request and verify the full artifact archive.

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